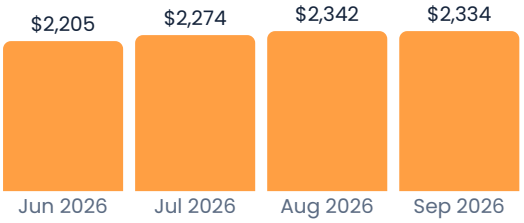
AWS Review + Fix Guide — Acme Shop (sample)

October 30, 2026 · 19 findings · read-only review

André Luís Pizzetti
AWS Certified Solutions Architect

Last full month bill	Estimated monthly savings	Estimated yearly savings	Security / reliability risks
$2,334	**$743**	**$8,910**	**10**

Monthly bill



Findings, in order of priority

#	FINDING	AREA	EST. SAVINGS / MO	PRIORITY
1	No Savings Plan for your steady compute usage	Cost	$177	High
2	2 EC2 instance(s) are oversized for their workload	Cost	$140	High
3	1 EC2 instance(s) almost idle (CPU under 5% the whole period)	Cost	$68	High
4	Root account has no MFA	Security	—	High
5	1 database(s) reachable from the internet	Security	—	High
6	1 database(s) without automated backups	Reliability	—	High
7	Old EBS snapshots that are probably no longer needed	Cost	$190	Medium
8	4 EBS volume(s) not attached to any server	Cost	$108	Medium
9	1 load balancer(s) with no servers behind them	Cost	$16	Medium
10	2 console user(s) without MFA	Security	—	Medium
11	2 S3 bucket(s) without Block Public Access	Security	—	Medium
12	No active CloudTrail (audit log)	Security	—	Medium
13	1 production database(s) in a single Availability Zone	Reliability	—	Medium

14	2 large S3 bucket(s) with no lifecycle rule	Cost	$29	Low
15	2 Elastic IP(s) reserved but not in use	Cost	$7	Low
16	1 instance(s) on older generations (t2/m4/c4/r4)	Cost	$7	Low
17	1 access key(s) older than 90 days	Security	—	Low
18	GuardDuty threat detection is off	Security	—	Low
19	1 database(s) with unencrypted storage	Security	—	Low

Savings are estimates based on list prices and the usage observed during the review period. Actual results depend on your workload and discounts. This review was prepared with the help of automated and AI-assisted analysis on anonymized data; every finding was checked by hand. No changes were made to your account.

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Details

1. No Savings Plan for your steady compute usage

High Cost · est. $177/month

A 1-year Compute Savings Plan typically gives 20-30% off EC2, Fargate and Lambda in exchange for a fixed hourly commitment. Committing ~70% of your lowest month keeps the risk low.

Affected resources

- Steady compute spend after the resizing above: about $1,012/month

How to fix it, step by step

1. Do the resizing and clean-up items first (they lower the baseline).
2. Console: Billing → Savings Plans → Recommendations → Compute Savings Plan, 1 year, No upfront.
3. Start with a commitment around $0.97/hour or lower; you can add more later.

2. 2 EC2 instance(s) are oversized for their workload

High Cost · est. $140/month

Average CPU stayed under 10% during the period, so a smaller (cheaper) size can run the same load.

Affected resources

- web-1 (m5.xlarge, us-east-1) — avg CPU 6.1%, max 31.0% → m5.large
- web-2 (m5.xlarge, us-east-1) — avg CPU 5.4%, max 28.5% → m5.large

How to fix it, step by step

1. Pick a low-traffic window and create an AMI of the instance (rollback point).
2. Console: EC2 → Instances → select → Instance state → Stop.
3. Actions → Instance settings → Change instance type → choose the suggested size → Start.
4. Watch CPU and memory in CloudWatch for 1 week. If CPU stays above 70%, go back one size.

3. 1 EC2 instance(s) almost idle (CPU under 5% the whole period)

High Cost · est. $68/month

These servers are running and billed every hour but do practically no work.

Affected resources

- old-staging (t2.large, us-east-1) — max CPU 2.9%

How to fix it, step by step

1. Confirm with the owner if the instance is still needed (check its Name tag and recent logins).
2. If not needed: create an AMI or snapshot as a backup, then stop it for 2 weeks.
3. If nobody complains, terminate it and delete the backup after 30 days.

4. Root account has no MFA

High Security

The root user can do anything, including closing the account. A leaked password is enough to lose it.

How to fix it, step by step

1. Sign in as root → account menu → Security credentials → Assign MFA device.
2. Use an authenticator app or a hardware key; store recovery codes safely.

5. 1 database(s) reachable from the internet

High Security

Databases should only accept connections from your application servers.

Affected resources

- reports-db (us-east-1)

How to fix it, step by step

1. RDS → database → Modify → Connectivity → Public access: No.
2. Check the security group allows the DB port only from the app's security group.

6. 1 database(s) without automated backups

High Reliability

If data is deleted or corrupted, there is no point-in-time restore.

Affected resources

- shop-db (us-east-1)

How to fix it, step by step

1. RDS → database → Modify → Backup retention period: 7 days (or more) → Apply immediately.

7. Old EBS snapshots that are probably no longer needed

Medium Cost · est. $190/month

Snapshots are incremental, so the real saving is usually lower than the estimate — but old ones add up.

Affected resources

- 38 snapshots older than 90 days in us-east-1 (3,800 GB)

How to fix it, step by step

1. Decide a retention rule (e.g. keep 30 daily + 12 monthly).
2. Use Amazon Data Lifecycle Manager or AWS Backup to apply it automatically.
3. Delete snapshots outside the rule (never the ones used by an AMI still in use).

8. 4 EBS volume(s) not attached to any server

Medium Cost · est. $108/month

Detached disks keep being billed every month even though nothing uses them.

Affected resources

- vol-0b0 (500 GB gp2, us-east-1)
- vol-0b1 (200 GB gp2, us-east-1)
- vol-0b2 (100 GB gp3, us-east-1)
- vol-0b3 (300 GB gp2, us-east-1)

How to fix it, step by step

1. Console: EC2 → Volumes → filter State = available.
2. For each volume: Actions → Create snapshot (cheap backup), then Actions → Delete volume.

9. 1 load balancer(s) with no servers behind them

Medium Cost · est. $16/month

A load balancer has a fixed hourly price even with zero traffic.

Affected resources

- old-promo-alb (application, us-east-1)

How to fix it, step by step

1. Confirm DNS records (Route 53 or your DNS provider) don't point to it.
2. Console: EC2 → Load Balancers → select → Actions → Delete.

10. 2 console user(s) without MFA

Medium Security

Users who log in to the console with only a password are the most common way accounts get compromised.

Affected resources

- USER-01
- USER-03

How to fix it, step by step

1. Ask each user to enable MFA (IAM → Users → Security credentials).
2. Optional: add a policy that denies actions when MFA is not present.

11. 2 S3 bucket(s) without Block Public Access

Medium Security

Without this setting a single wrong policy or ACL can expose files to the internet.

Affected resources

- acme-shop-logs
- acme-backups

How to fix it, step by step

1. If the bucket does not serve a public website: S3 → bucket → Permissions → Block public access → Edit → enable all four options.
2. Even better: enable it at account level (S3 → Block Public Access settings for this account).

12. No active CloudTrail (audit log)

Medium Security

Without CloudTrail there is no record of who changed what — essential after any incident.

How to fix it, step by step

1. CloudTrail → Create trail → apply to all regions → new S3 bucket. The first trail is free (you only pay S3 storage).

13. 1 production database(s) in a single Availability Zone

Medium Reliability

If that zone has a problem, the database is down until AWS recovers it.

Affected resources

- reports-db (us-east-1)

How to fix it, step by step

1. Decide if the business can accept a few hours of downtime.
2. If not: RDS → Modify → Multi-AZ deployment: Yes (roughly doubles the DB instance cost).

14. 2 large S3 bucket(s) with no lifecycle rule

Low Cost · est. $29/month

Data that is rarely read costs less in S3 Intelligent-Tiering or Glacier classes.

Affected resources

- acme-shop-logs (2,400 GB in Standard)
- acme-backups (1,800 GB in Standard)

How to fix it, step by step

1. S3 → bucket → Management → Create lifecycle rule.
2. Logs/backups: move to Glacier Instant Retrieval after 30 days, expire after 365 days.
3. Unknown access pattern: transition to Intelligent-Tiering after 0 days.

15. 2 Elastic IP(s) reserved but not in use

Low Cost · est. $7/month

AWS charges for every public IPv4 address, used or not.

Affected resources

- eipalloc-01 (us-east-1)
- eipalloc-02 (us-east-1)

How to fix it, step by step

1. Console: EC2 → Elastic IPs → select the unassociated address → Actions → Release.

16. 1 instance(s) on older generations (t2/m4/c4/r4)

Low Cost · est. $7/month

Newer generations (t3, m6i, c6i, r6i) cost about the same or less and are faster.

Affected resources

- old-staging (t2.large, us-east-1)

How to fix it, step by step

1. Same procedure as a resize: AMI → stop → change instance type (e.g. t2.medium → t3.medium) → start.
2. Check the AMI supports the Nitro system (ENA and NVMe drivers) before changing.

17. 1 access key(s) older than 90 days

Low Security

Long-lived keys are often forgotten in laptops, scripts and repositories.

Affected resources

- USER-01 (key 1, last rotated 2024-03-02)

How to fix it, step by step

1. Create a new key, update the application, deactivate the old key, wait 1 week, delete it.
2. Better: use IAM roles (EC2 instance profiles, ECS task roles) instead of keys.

18. GuardDuty threat detection is off

Low Security

GuardDuty spots crypto-mining, leaked keys and unusual logins; it usually costs a few dollars/month.

Affected resources

- us-east-1

How to fix it, step by step

1. GuardDuty → Get started → Enable (30-day free trial shows the real cost).

19. 1 database(s) with unencrypted storage

Low Security

Encryption at rest is required by most compliance frameworks and is free.

Affected resources

- reports-db (us-east-1)

How to fix it, step by step

1. Encryption can't be turned on in place: take a snapshot → copy it with encryption → restore.
2. Plan a short maintenance window.